# 04 Access Techs

04.01-04.04 Access Networks, xDSL, Optical Networks, Other access types.

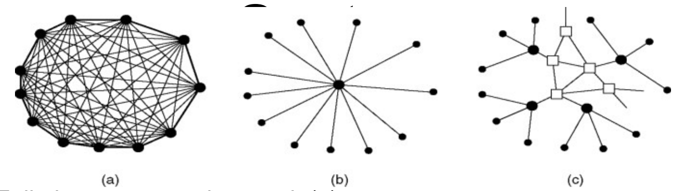




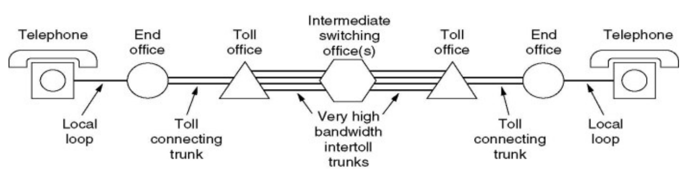
## Public Switched Telephone System

- Structure of the Telephone System
  - Why do we need to study PSTN? Because it's one of the main infrastructures strictly interleaved to computer networks
  - We will analyze:
    - The Local Loop: Modems, ADSL and Wireless
    - Trunks and Multiplexing
    - Switching

# Structure of the Telephone




- Fully-interconnected network (a)
  - Absolutely inadequate since first telephone days
- Centralized switch (b)
  - First solution: introduce one hierarchy level
- Multi-level hierarchy (2 levels in fig.(c))
  - Introduce more hierarchy levels




A typical circuit route for a medium-distance call.





# Major Components of the Telephone System

- Local loops
  - **Analog** twisted pairs going to houses and businesses
- Trunks
  - **Digital** fiber optics connecting the switching offices
- Switching offices
  - Where calls are moved from one trunk to another

- **Analog** vs. **digital**
  - Reconstruction/re-generation of signals
  - Signal/Noise ratios
  - Better reliability of digital
  - All these issues make digital also more economic

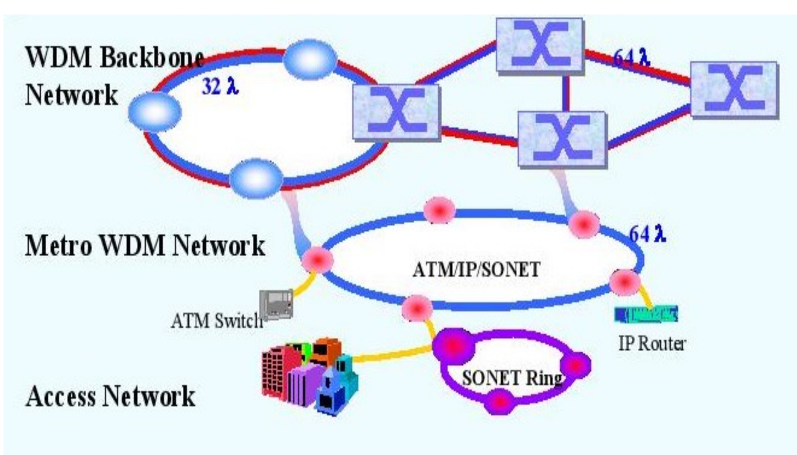
# Architectures Of Networks

- Backbone Networks
  - networks in the same building, in different buildings in a campus environment, or over wide areas.
  - exchange of information between different LANs
- Metro Area Networks (MAN)
  - network that interconnects users with computer resources in a geographic area or region larger than that covered by even a large local area network (LAN) but smaller than the area covered by a wide area network (WAN).
- **Access Networks**
  - Distributed EPON architectures
  - Distributed ring-based WDM-PON architectures
  - Converged Optical/Wireless Access Networks





# Architecture Of Networks



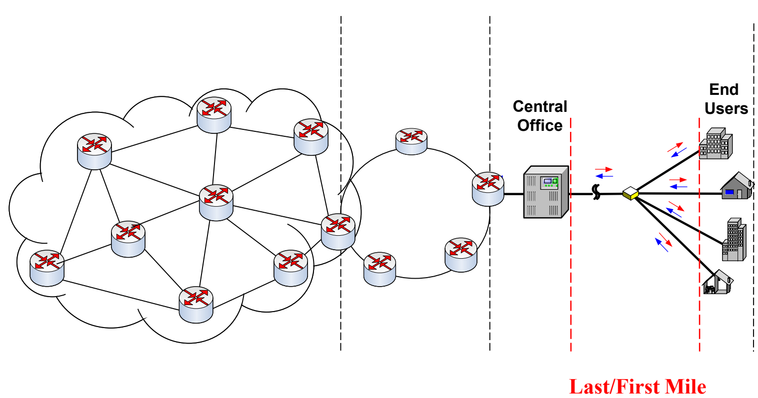
# Access Networks : Introduction

- Access Networks
  - Tremendous growth in both backbone and Metro Access Network (MAN) capacity.
  - End users are becoming more sophisticated
    - Rich multimedia
    - Real-time services
  - **The "Last Mile" remains a bottleneck.**
    - Current "*Last Mile"* capacity has increased from 56Kb/s (dialup modem) to a few Mb/s (cable modem or digital subscriber line (DSL) connection).
    - Still far short of the **Gigabit line speed** necessary to support rich multimedia and real-time services.





# Access Networks ...



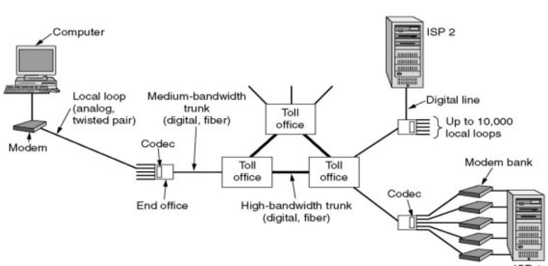
# Access Networks …

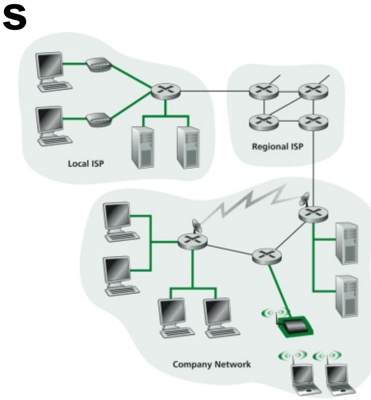
- Copper-based access networks will soon no longer be able to meet the ever-growing consumer demand for bandwidth.
- PON-based fiber-to-the-curb/home (FTTC/FTTH) systems are considered as possible successors to current copper-based access solutions.
- Two most viable architectures:
  - Single channel Time-Division Multiplexed PON (TDM-PON)
  - Multi-channel Wavelength-Division Multiplexed PON (WDM-PON)





# The Local Loop: Modems, ADSL, and Wireless



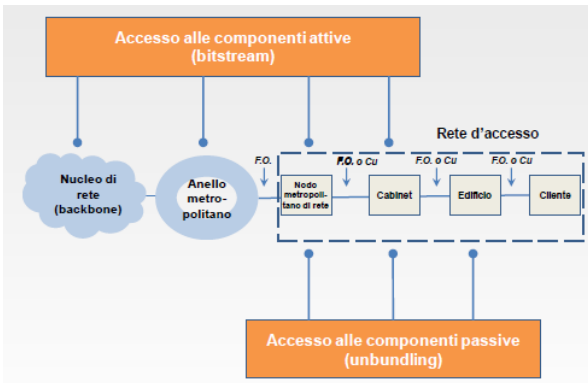






Access networks

- The use of both analog and digital transmissions for a computer to computer call. Conversion is done by the modems and codecs.
- Problems in communications:
  - Attenuation (as a function of frequency)
  - Distortion (different frequencies travel with different speeds)
  - Noise (thermal, impulsive)
- As a consequence Base-band transmissions can only be used at low speeds and distances

# Wired Access




- Main elements of a wired network for Internet access





# Modems and (de)modulation

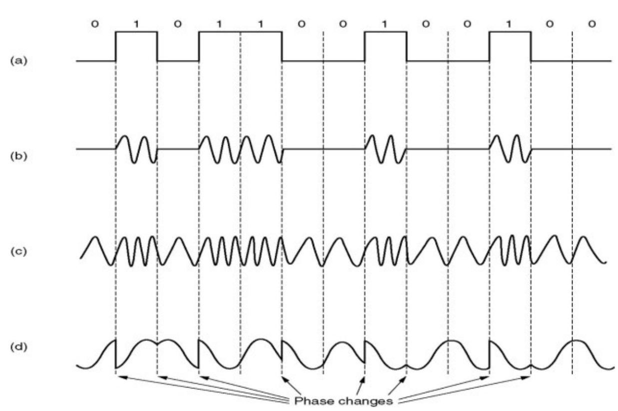
- A sinusoidal signal (*carrier*) (f = 1000-2000 Hz) is modulated, i.e. one of its characteristics (ampl., freq., phase) is modified accordingly to the information signal (*modulating signal*) to produce the *modulated signal*
- Frequency modulation is achieved by using 2 different tones (carriers)

Sample Binary signal

Amplitude modulation (ASK (Amplitude Shift Keying))

Frequency modulation (FSK (Frequency Shift Keying))

Phase modulation (PSK(Phase Shift Keying))



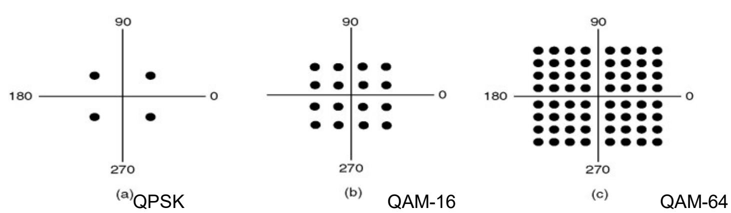
# Modulation: some definitions and relations

- Pass bandwidth: frequency interval that transit along the means of transmission with minimum attenuation
- Baud = Nr. of samples transmitted per second. 1 baud→ 1symbol (symbol rate = bauds)
- Bit rate: information amount transmitted along the channel (modulation techniques determine the nr. of bits per symbol) every time unit → = symbol rate x bits per symbol



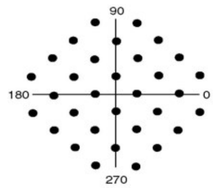


# Modems: constellation diagrams

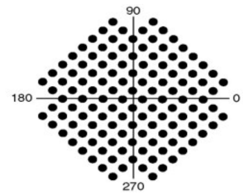



- Constellation diagrams show the admissible combination of amplitude and phase. Each modem has a proper c.d.
- QPSK: 4 valid combinations, 2 bits/symbol
- QAM-16: 16 different combinations, 4 bits/symbol (2400 baud line → 9600 bps)
- QAM-64: 64 different combinations, 6 bits/symbol

# Modems: constellation diagrams




a) V.32 for 9600 bps




b) V.32 bis for 14400 bps

- Error correction: extra bits added to each sample (TCM = Trellis Code Modulation)
- V.32 : 32 points in c.d., 4 + 1 (parity) bits, 2400 bauds → 9600 bps + error correction
- V.32 bis: 32768 points in c.d., 14 +1 bits/symbol, 2400 bauds → 33600 bps
- Standard modems: 33.6 kbps, but …
  - Phone bandwidth = 4 kHz → 8 ksamples/s, (7+1) bit/sample → 56kbps → V.90 standard



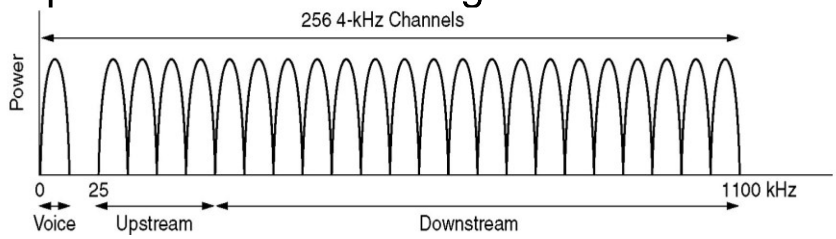


# ADSL: Asymmetric Digital Subscriber Line

- ADSL uses a frequency spectrum of 1.1 MHz. Divides it into 256 channels each of size roughly 4312.5 Hz.
- Channel 0 : POTS
- Channels 1-5 ; guard band between voice and data
- Two for control channels, one for downstream and one for upstream
- Remaining are partitioned between upstream and downstream : depends on the service provider; usually it is asymmetric giving 80-90% for download and remaining for upstream – hence the word Asymmetric

# Digital Subscriber Lines

Operation of ADSL using discrete multitone





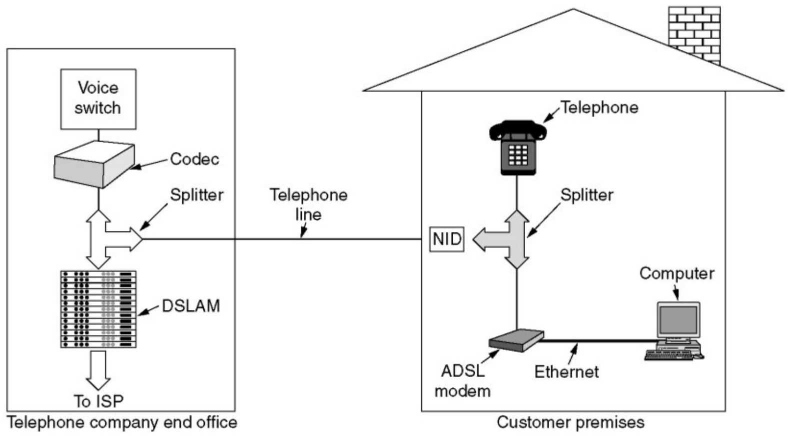



# ADSL

- Within each channel, modulation scheme similar to V.34 is used ;
- QAM with 15 bits per baud
- 4000 baud instead of 2400
- With 224 downstream channels, download speed 13.44 Mbps is theoretically possible
- In practice, S/N ratio is never good enough to achieve this rate, but 8 Mbps is possible on short runs over high quality local loops

# Installation requirement of ADSL

A typical ADSL equipment configuration.





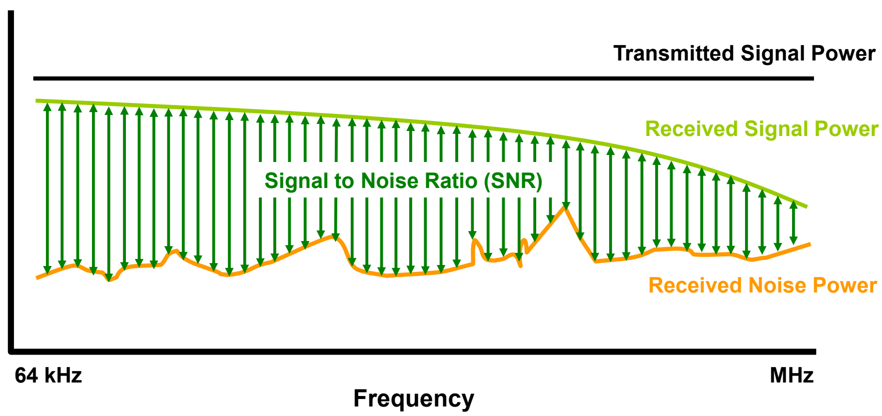



# VDSL

- Very-high-bit-rate digital subscriber line (VDSL) are digital subscriber line (DSL) technologies providing data transmission faster than asymmetric digital subscriber line (ADSL).
- VDSL offers speeds of up to 52 Mbit/s downstream and 16 Mbit/s upstream, over twisted pair of copper wires using the frequency band from 25 kHz to 12 MHz.
- These rates mean that VDSL is capable of supporting applications such as high-definition television, as well as telephone services (voice over IP) and general Internet access, over a single connection.

# Signal Attenuation

***SNR is responsible for the performance***

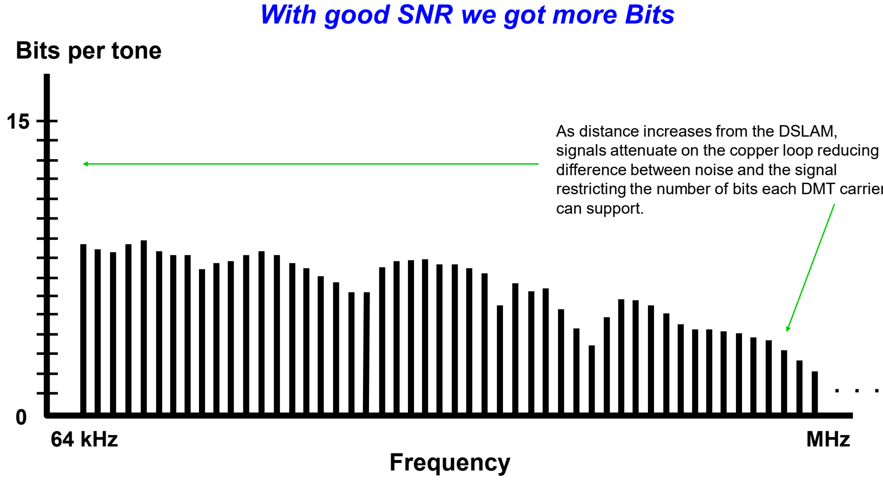







# Bits per Tone

***With good SNR we got more Bits***



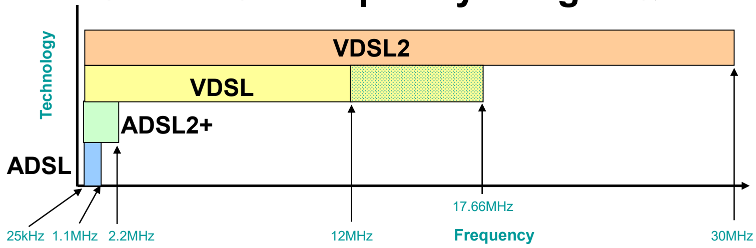
# Standards

| Family | ITU | Name | Ratified | Maximum Speed capabilities |
|---|---|---|---|---|
| ADSL | G.992.1 | G.dmt | 1999 | 8 Mbps down<br>800 kbps up |
| ADSL2 | G.992.3 | G.dmt.bis | 2002 | 8 Mb/s down<br>1 Mbps up |
| ADSL2plus | G.992.5 | ADSL2plus | 2003 | 24 Mbps down<br>(Amend 1, 29Mbps)<br>1 Mbps up |
| ADSL2-RE | G.992.3 | Reach Extended | 2003 | 8 Mbps down<br>1 Mbps up |
| **VDSL** | **G.993.1** | **Very-high-data-rate DSL** | **2004** | **55 Mbps down**<br>**15 Mbps up** |
| **VDSL1 -12 MHz long reach** | **G.993.2** | **Very-high-data-rate DSL 2** | **2005** | **55 Mbps down**<br>**30 Mbps up** |
| **VDSL2 - 30 MHz Short reach** | **G.993.2** | **Very-high-data-rate DSL 2** | **2005** | **100 Mbps up/down** |





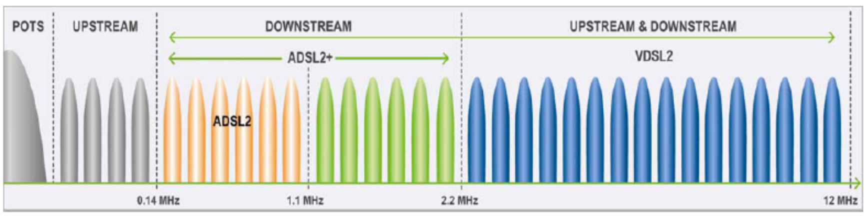
# ADSL - VDSL Frequency Ranges & Rates




| Technology | Freq range | Max Rates | Max # of carriers and Bin spacing |
|---|---|---|---|
| ADSL | 25kHz – 1.1MHz | 800kbps up<br>8Mbps down | 256 with 4.3125kHz bins |
| ADSL2+ | 25kHz – 2.2MHz | 1Mbps up<br>24Mbps down | 512 with 4.3125kHz bins<br>Amend. 1 = 29 Mbps down |
| VDSL(1) | 25kHz – 12MHz | 15Mbps up<br>55Mbps down | 2782 with 4.3125kHz bins |
| VDSL2 | 25khz – 30MHz | 100Mbps up<br>100Mbps down | 4096 with 4.3125kHz bins<br>3478 with 8.625kHz bins |

# VDSL Standards

| Version | Standard name | Common name | Downstream rate | Upstream rate | Approved on |
|---|---|---|---|---|---|
| VDSL | ITU G.993.1 | VDSL | 55 Mbit/s | 3 Mbit/s | 2001-11-29 |
| VDSL2 | ITU G.993.2 | VDSL2 | 100 Mbit/s | 100 Mbit/s | 2006-02-17 |
| VDSL2-Vplus | ITU G.993.2 Amendment 1 (11/15) | VDSL2 Annex Q VPlus/35b | 300 Mbit/s | 100 Mbit/s | 2015-11-06 |








# Optical Networks

- **Definition:** An Optical Network is a telecommunication network
  - with transmission links that are optical fibers and
  - with an architecture that use designed to exploit the unique features of fibers.
- High performance lightwave network –involve complex combination both optical and electronic devices.
- Low-cost broadband services – Internet based applications continues to increase.
- The “glue” that holds the purely optical network together consists of :
  - optical network nodes (ONN) connecting the fibers within the network
  - network access stations (NAS) interfacing user terminals and other non- optical end systems to the network
- **Critical role :**
  - Reducing communications costs
  - Promoting competition among carriers & service providers
  - Increasing the demand for new services

# Why Optical Networks ...??

**Dramatic changes in the telecommunication industry.**

- Need for more capacity in the network.
- Tremendous growth of the Internet and the World Wide Web in terms of
  - number of users & the amount of time
  - bandwidth taken by each user – internet traffic growing rapidly.
- Businesses rely on high speed networks.
- Need for more bandwidth.
- Deregulation of the telephone industry.
- Need of providing quality of service(QoS) to carry performance sensitive applications ( real-time voice, video etc.)





# Generations of Optical Networks

- **First Generation:**
  - Optics used for transmission & provide capacity
  - Switching & other intelligent network functions were handled by electronics
    - ex. SONET (synchronous optical network)
    - SDH ( synchronous digital hierarchy)
- **Second Generation:**
  - have routing ,switching and intelligence in the optical layer
  - use multiplexing techniques – provide the capacity needed

## The Optical Layer
## Layered View of the Optical Network

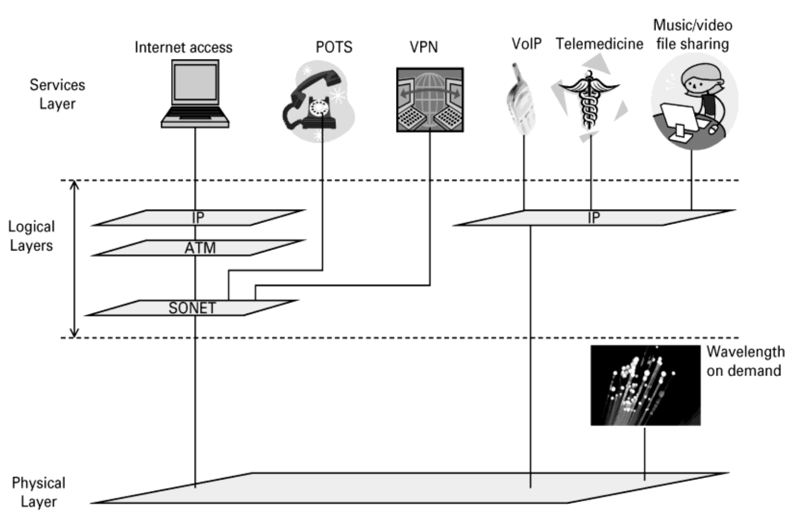
The architecture is composed of an underlying **optical infrastructure**

- Physical layer
  - Contains optical components executing linear(transparent) operations on optical signal.
  - provides basic communication services to a number of independent logical networks (LNs).
- LNs are residing in the Data Link layer.
  - Contains electronic components executing nonlinear operations on electrical signal





## The Optical Layer
## Layered View of the Optical Network



# Functions Of The Optical Layer

- Multiplexes lightpaths into a single fiber.
- Allows individual lightpaths to be extracted efficiently from the composite multiplex signal at the network nodes.
- Incorporates sophisticated service restoration techniques.
- Incorporates management techniques.
- Provides lightpaths – used by SONET and IP network elements.





# Advantages of Layering

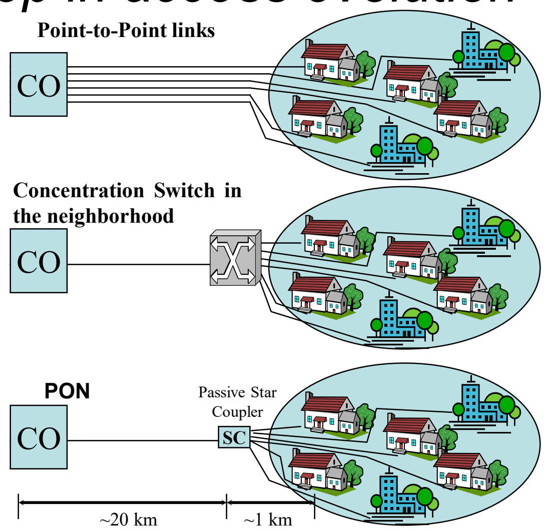
1. Independently control and manage each logical network →simplifying these functions.
2. Share the total resources of the physical layer among several logical network → exploiting them more efficiently.
3. Customize each logical network to provide specialized user services → improving the QoS.
4. Dynamically reconfigure each logical network → equipment failures and changing traffic patterns.
5. Use both optical and electronic degrees of freedom→ provide flexibility, survivability, manageability and capacity for growth and change.

# Why Passive Optical Networks?

## *A natural step in access evolution*

**PON**

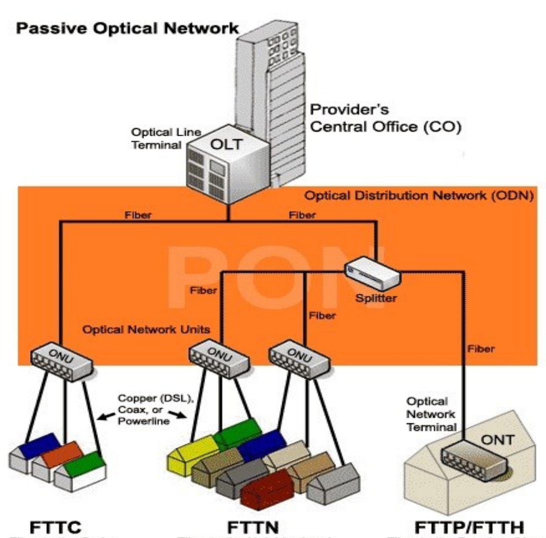
- Minimum fiber usage/ N+1 transceivers
- Path transparency
- Passive network elements
- Much longer distance (~20km) than DSL (~5.5 km).
- Higher bandwidth due to deeper fiber penetration.
- Downstream video broadcasting.



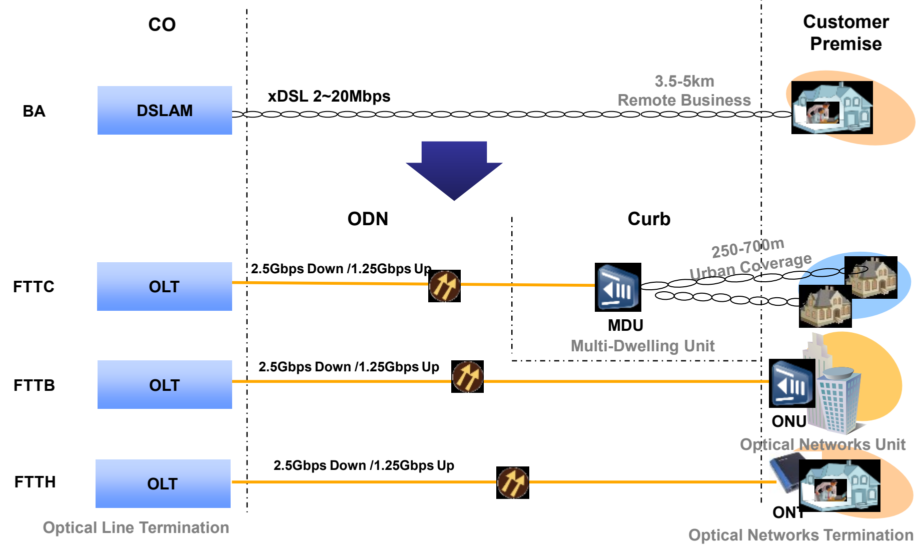





# Passive Optical Access Network



# Architecture of Optical Access Network

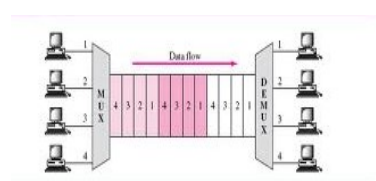




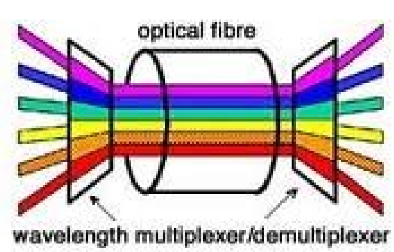



# Multiplexing Techniques

Time Division Multiplexing (TDM) :
A type of multiplexing that combines data streams by assigning each stream a different time slot in a set. TDM repeatedly transmits a fixed sequence of time slots over a single transmission channel.




Wavelength Division Multiplexing (WDM):
A technique of sending signals of several different wavelengths of Light into the Fiber simultaneously. In fiber optic communications, wavelength-division Multiplexing (WDM) is a technology which multiplexes multiple optical carrier signals on a single Optical Fiber by using different wavelengths (colors) of Laser light to carry different signals.WDM is similar to frequency-division multiplexing (FDM).



# Optical Network Terminal and Optical Network Unit

**ONT (Optical Network Terminal):**

•An ONT is a media converter that is installed either outside or inside your premises, during fiber installations.

•The ONT converts fiber-optic light signals to copper/electric signals.

•Three wavelengths of light are used between the ONT and the Optical Line Terminal :
•1310 nm voice/data transmit
•1490 nm voice/data receive
•1550 nm video receive

Each ONT is capable of delivering:
• Multiple POTS (plain old telephone service) lines
• Internet data
•Video

**ONU (Optical Network Unit):**

•An Optical Network Unit (ONU) converts optical signals transmitted via fiber to electrical signals.

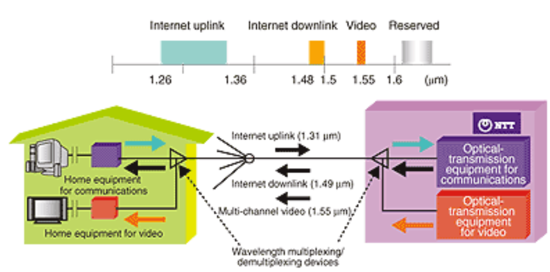
•These electrical signals are then sent to individual subscribers. ONUs are commonly used in fiber-to-the-home (FTTH) or fiber-to-the-curb (FTTC) applications.





# Transmission between ONT and ONU Example …

•Using different wavelengths for each service makes it possible to transmit high-speed Internet and video services at the same time.
• The 1310nm and 1490nm bands are used for Internet transmissions on the uplink and downlink, respectively,
•The 1550nm band is used for multi-channel video broadcasts.
•Wavelength multiplexing is performed at the central office and a wavelength demultiplexing mechanism is provided at the customer's house.



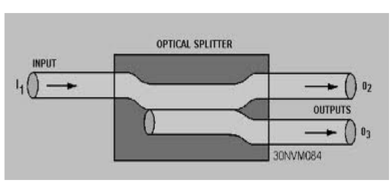
# OLT – Optical Line Terminal

- OLTs are located in provider's central switching office.
- This equipment serves as the point of origination for FTTP (Fiber-to-the-Premises) transmissions coming into and out of the national provider's network.
- An OLT, is where the PON cards reside. The OLT's also contain the CPU and the GWR and VGW uplink cards. Each OLT can have a few or many dozens of PON cards.
- PON = Passive Optical Network
  GWR = Gateway Router
  VGW = Voice Gateway
- Each PON card transmits 1490nm laser data signal to the ONT, and receives the ONT transmission of the 1310nm laser data signal.
- The one-way 1550nm laser video signal to the ONT is injected into the fiber at the CO.

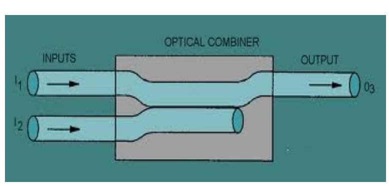




# Optical Splitter and Combiner



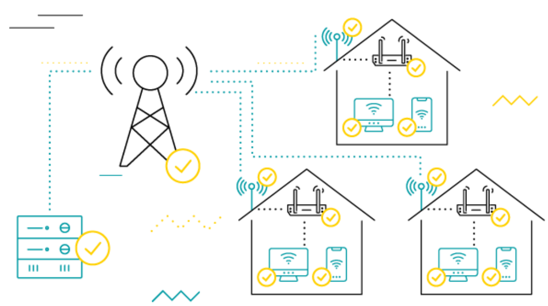

Fiber optic splitter is used to split the fiber optic light into several parts at a certain ratio. For example, a 1X2 50:50 fiber optic splitter will split a fiber optic light beam into two parts, each get 50 percent of the original beam.




An optical combiner is a passive device that combines the optical power carried by two input fibers into a single output fiber.

# FIXED WIRELESS ACCESS

- What is FWA?
  - Broadband, Broaderband, Narrowband, Voice, Data, INTERNET, Video, Telemedicine, Tele-Education, Connectivity, . . .
- Data over FWA; MegaBits and even, sometime, GigaBits/Second
- FWA is not an allocation or spectrum designation







# FIXED WIRELESS ACCESS

- FWA works in multiple bands
- There is a diversity of FWA needs and services
- No one standard because FWA customers don’t move around
- Technology becomes cheaper in the marketplace

# FWA VISION

- Promote Competition
- Deregulate as competition develops
- Protect consumers
- Ensure broad access to communications services and technology
- Advance competitive goals worldwide



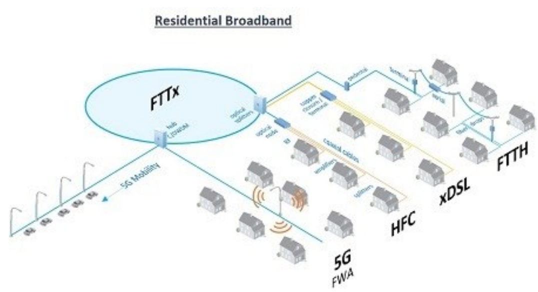


# FWA FACTORS

- Need to transmit larger volumes of information
- Increased spending by small and mid-sized business
- Integrate voice and data but it's more difficult then wired
- Need for greater interoperability
- A requirement for cost-effective solutions to business problems

# Network Convergence

- The primary benefit of converged infrastructure is to scale and bring new services to market rapidly.








# SpaceX's Starlink

- SpaceX's Starlink system, is a constellation of interlinked satellites
- It is intended to provide high-speed internet to underserved rural areas.
- As of June 2021, there are over 1,500 Starlink active satellites, making Starlink the largest satellite constellation around Earth.
- Starlink satellites reside into low-Earth orbit

# How do satellite constellations like Starlink work

- Satellite internet can be notoriously laggy.
  - Starlink satellites occupy much lower orbits than traditional satellites (only 550 kilometers)
- Starlink is expected to deliver speeds up to about one gigabit per second.
  - Now data speeds vary from 50 to 150 megabits per second and latency from 20 to 40 milliseconds depending on Earth zones.
- Starlink map





# Starlink CPE

- All you need to do to make the connection is set up a small satellite dish at home
- The received signal pass the bandwidth on to router.
- There's a Starlink app for Android and iOS to help customers pick the best location and position for their receivers.
- There are problems with “thermal shutdown” once dishes hit 50° Celsius

# Low Earth Orbit Visualization

- Low Earth Orbit Visualization